

# Text Classification in NLP

CMPE346 — Group 2

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**Group task:** Text Classification

**Metric:** Accuracy & Weighted F1

| Student     | Task                         |
|-------------|------------------------------|
| Atakan Gül  | Media Bias Detection in News |
| [Student 2] | [Topic]                      |
| [Student 3] | [Topic]                      |

## BiLSTM

Bidirectional LSTM trained from scratch. Captures sequential context in both directions.

## BERT

Transformer encoder pretrained via masked language modeling. Fine-tuned end-to-end. `bert-base-uncased`.

## RoBERTa

Optimized BERT: removes next sentence prediction, uses dynamic masking and more data. `roberta-base`.

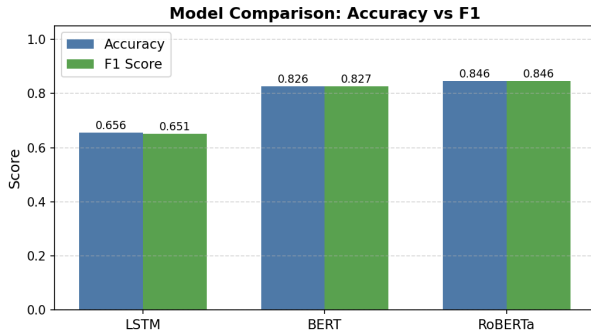
**Task:** Classify a news sentence as *biased* or *not biased*.

**Dataset: BABE** (Bias Annotations By Experts)

- ▶ 4,121 expert-annotated news sentences
- ▶ Diverse outlets, binary labels
- ▶ 3,121 train / 1,000 test

| Model   | Acc.         | F1           |
|---------|--------------|--------------|
| BiLSTM  | 0.656        | 0.651        |
| BERT    | 0.826        | 0.827        |
| RoBERTa | <b>0.846</b> | <b>0.846</b> |

RoBERTa best. +19% over BiLSTM.



**Task:** [Description]

**Dataset:** [Name — brief stats]

| Model   | Acc. | F1 |
|---------|------|----|
| BiLSTM  | —    | —  |
| BERT    | —    | —  |
| RoBERTa | —    | —  |

*[Student 2 key finding]*

**Task:** [Description]

**Dataset:** [Name — brief stats]

| Model   | Acc. | F1 |
|---------|------|----|
| BiLSTM  | —    | —  |
| BERT    | —    | —  |
| RoBERTa | —    | —  |

*[Student 3 key finding]*

# Conclusion

- ▶ RoBERTa outperforms BERT and BiLSTM across all three tasks
- ▶ Pretraining is decisive on small datasets
- ▶ Future work: domain-adaptive pretraining, larger model variants